CWBSNet: A Segmentation Network for Comple Water Bodies in Remote Sensing Images

Piaoling Xu, Jiguang Dai, Tengda Zhang, and Yujie Wu

***Abstract*— To address the challenges posed by diverse water body morphologies, irregular boundaries, and significant reflectance variations in remote sensing images—factors often overlooked by general-purpose models that lead to local omissions, boundary mislocalizations, and false detections—we propose a novel complex water body segmentation network (CWBSNet). This network is designed for water body extraction tasks on both the Sentinel-2 dataset S1S2-Water and the aerial dataset FLAIR#1. CWBSNet employs a transformer-based backbone to generate four-level feature embeddings with long-range global dependencies. To enhance global water body representation and mitigate local omission issues, we introduce the spectrum transform module (STM), which exchanges spectral information between water and nonwater regions on small-scale features via frequency-domain transformations. Furthermore, we design a multilevel feature extraction and fusion module (MFEFM), comprising Block1, a spatial-channel information fusion block (SIFB), and Block3, which enhances local feature representations through attention mechanisms, multiscale convolutions, and spatial-channel transformations. Directional convolutions are incorporated into shallow features to capture orientation cues and refine boundary detail representation. In addition, based on the strong response of near-infrared (NIR) and normalized difference water index (NDWI) to water bodies, we propose the information augmentation module (IAM). This module integrates rotary position embedding (ROPE) attention into a linear attention transformer to support water body recognition and reduce false positives. Finally, a decoder generates the saliency map. Experiments on two public datasets demonstrate that CWBSNet achieves superior performance in both training efficiency and detection accuracy compared to the state-of-the-art methods. The code is available at: https://github.com/Lingxp/CWBSNet**

***Index Terms*— High-resolution remote sensing image, normalized difference water index (NDWI), spectral signature, transformer, water segmentation.**

I. Introduction

WATER resources are fundamental to maintain the Earth's ecosystem and directly influence the stability of the ecological environment, playing a crucial role in climate regulation and ecological cycles [1]. Proper allocation and effective management of water resources are essential for addressing extreme weather events and mitigating ecological degradation. Therefore, rapid and accurate identification of water bodies is significant for water quality monitoring [2], flood detection [3], climate change [6], and land use planning [5]. Traditional field data collection methods struggle to meet the demand for large-scale, real-time water body data across various sectors. In contrast, remote sensing imagery, with its extensive coverage and short revisit periods, has become a primary means for timely acquisition of water body distribution information [7], [8].

Compared to traditional water body extraction methods, such as single-band thresholding [59], spectral relationship methods [60], and spectral classification methods [61], deep learning-based extraction approaches demonstrate better accuracy and generalization capabilities, enabling large-scale automated extraction [10], [62], [63]. Variations in water reflectance due to lighting effects, the complexity of natural water body boundaries, and the diversity of water body types are fundamental reasons for issues, such as water body disconnection, blurred boundaries, and extraction omissions. Existing methods primarily address these challenges by capturing multiscale features [11], spatial features [12], and spectral features [13]. While these methods can achieve excellent performance in large water bodies or scenarios with minimal reflectance variation, they struggle when multiple water types with distinct boundary characteristics and varying reflectance coexist in remote sensing images. General models are based on the fundamental assumption that water bodies possess smooth textures and uniform light reflection. However, this assumption fails for remote sensing images containing diverse water forms. The images display a variety of water types, including rivers, lakes, ponds, wetlands, and reservoirs. In Fig. 1, in image group Fig. 1(a), the same water body, influenced by different lighting, cannot accurately represent its spectral characteristics within the shaded area of the red box, while the bright area in the yellow box may lead to strong surface reflections. The poor performance of general segmentation algorithms in these situations stems from the variable texture characteristics that hinder targeted learning. In image group Fig. 1(b), unlike the assumed water bodies, edges along lakes, rivers, or wetlands may exhibit numerous twists and branches. The flow rate and complex terrain further influence the shape and smoothness of the water surface, leading to more irregular and unclear boundaries, as seen in the purple box. Image group Fig. 1(c) showcases diverse water body features, clearly indicating different representations of the water bodies within the green and pink boxes under varying remote sensing conditions, illustrating the strong spectral heterogeneity of water bodies. Consequently, when the existing methods

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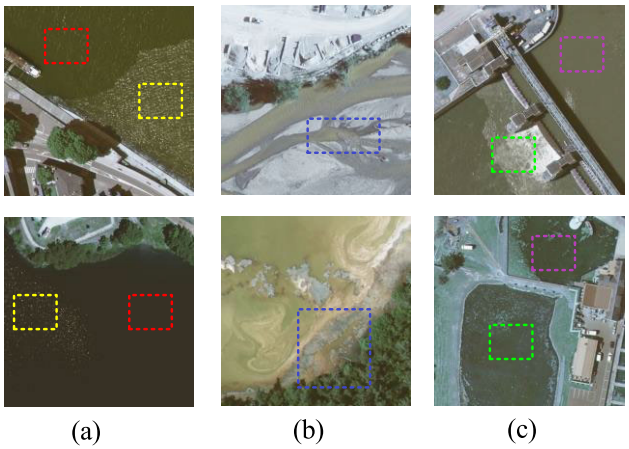


Fig. 1. Motivation for proposing CWBSNet. (a) Different water reflectivities. (b) Obscure boundary. (c) Diversity of water body.

are applied to urban remote sensing satellite images with multiple water types, the extraction results often suffer from significant water body disconnection and inaccurate boundary localization, complicating the application of remote sensing imagery.

To address the aforementioned issues, this article proposes complex water body segmentation network (CWBSNet) for the precise extraction of water bodies in images. The phenomenon of varying texture features reflected by the same water body makes it more reliable to identify these features using a combination mechanism of local and global contours. Based on the significant spectral variation between water bodies and land areas, we introduce the spectrum transform module (STM), which enhances the overall feature representation of water bodies by exchanging and fusing spectral information from deep features, in conjunction with spatial convolution. Due to the highly variable boundary rules and diverse lighting reflections that disrupt the coherence of simple water features, the encoder struggles to capture sufficient water body characteristics. While the encoder can recover extensive water body features, accurately restoring boundary details is challenging for conventional convolutional learning operations. To address this, we designed the multilevel feature extraction and fusion module (MFEFM), which learns the rules of water body boundaries by integrating multiscale spatial and channel features. This module employs directional convolution operations, multiscale feature processing, and self-attention mechanisms to enhance the perception of water body boundaries. Moreover, despite the diversity of water bodies, water exhibits unique reflectance characteristics across different spectral bands [such as visible light and near-infrared (NIR)] as well as in the normalized difference water index (NDWI). Our encoder incorporates the information augmentation module (IAM), which utilizes position encoding and linear attention mechanisms in these band mappings and further extracts and processes NIR and NDWI features through convolutional and fully connected layers.

We propose a novel network named CWBSNet, and the main contributions of this study are summarized as follows.

1) Complex waterbodies often exhibit abrupt reflectance changes, resulting in fragmented segments within homogeneous water areas and partial detection omissions. These challenges are further exacerbated by conventional models' limited receptive fields and lack of global context. To address these limitations, we propose a novel module that explicitly enhances continuity representation for regions with spectral discontinuities through global spectral modeling of fine-scale features.
2) Water body boundaries are frequently affected by terrain and anthropogenic disturbances, forming irregular multiscale geometric structures that blur the localization of edges. Conventional pooling operations or large convolution strides lead to the loss of fine details during downsampling. To mitigate this, we introduce MFEFM that reinforces boundary detail representation via hierarchical feature fusion. Moreover, it dynamically optimizes the semantic confidence of edge regions, thereby improving the spatial precision of complex boundary localization.
3) The NIR and NDWI bands exhibit strong absorption characteristics for waterbodies, which is particularly beneficial for distinguishing water and nonwater regions in complex morphological scenarios. However, general-purpose models often overlook the specific information from the NIR band. To bridge this gap, we propose IAM that collaboratively models the water-responsive features from the NIR and NDWI bands.

To validate the above contributions, we evaluate the proposed method on two public datasets: the Sentinel-2-based S1S2-Water dataset (10 m resolution) [14] and the airborne dataset FLAIR #1 (0.2-m resolution) [15], both of which contain water regions with irregular boundaries and complex reflectance characteristics—challenges typically underrepresented in the existing waterbody datasets. Experimental results demonstrate that CWBSNet effectively captures diverse waterbody structures and accurately delineates boundaries, achieving robust performance across remote sensing images of varying resolutions.

The remainder of this article is organized as follows. Section II reviews related work on waterbody extraction. Section III presents the proposed CWBSNet in detail. Section IV introduces the datasets and experimental setup and presents both comparative and ablation studies. Finally, Section V concludes this article.

II. Related Work

Over the past few decades, numerous methods have been proposed for extracting water bodies from remote sensing images. Unlike traditional approaches (such as histogram of oriented gradients [64], local binary patterns [20], scale-invariant feature transform [21], and support vector machines [22]), deep learning methods have undergone extensive research and have refined water body extraction networks from various perspectives. These methods can be broadly analyzed from two angles: 1) water body extraction methods based on different imaging techniques and 2) segmentation methods based on model structure optimization.

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Remote sensing images can be categorized into synthetic aperture radar (SAR) images, multitemporal images, hyperspectral images, multispectral images, high-resolution images, and aerial images. In the case of SAR images, Peña et al. [46] utilized the NDWI as a teacher model to train a convolutional neural network (CNN), demonstrating that the DEEPAQuA method better segments water bodies from SAR images. For multitemporal images, Landsat imagery is most commonly used for segmentation extraction. Wang et al. [25] proposed combining Google Earth Engine and multiscale CNNs for online predictions of urban water bodies from Landsat images. O'Sullivan et al. [26] proved that the U-Net model can effectively segment coastal water bodies in the Landsat dataset. Hyperspectral images contain rich spectral information in each pixel, including those from Sentinel-2 and high-resolution images. O'Sullivan et al. [45] explored the importance of NIR spectral bands for extracting coastal water bodies from Sentinel-2 images using an adapted U-Net architecture. Barbato et al. [28] proposed an unsupervised hyperspectral image water body segmentation method that does not require neural models but instead utilizes handcrafted features and improved linear iterative clustering to reduce noise impact on hyperspectral images. Multispectral images, which have smaller data volumes and faster processing speeds, are suitable for routine monitoring and rapid analysis applications. Panuntun et al. [27] demonstrated that deep learning semantic segmentation yields the best water body segmentation results in multispectral images compared to hyperspectral and aerial imagery. High-resolution images are the most commonly used objects in remote sensing extraction and segmentation. Onim et al. [29] proposed a time- and memory-efficient water body classification method to replace labor-intensive approaches and related costs, analyzing how fast fully convolutional networks (FastFCNs) perform semantic segmentation on high-resolution satellite images. Polushko et al. [47] created the flood dataset BlessemFlood21 and concluded through training semantic segmentation models that the dataset requires not only high-resolution RGB data but also NIR information. Additionally, aerial images generally have higher ground resolution among remote sensing images, making them more suitable for fine segmentation. Zhu et al. [32] proposed a new weighted mask loss function to address the class distribution imbalance in classification tasks for aerial images, accurately predicting water bodies. The Resunet, proposed by Diakogiannis et al. [65] combined with the dice loss function, significantly improves the performance of aerial image segmentation. Jonnalagadda and Hashim [33] introduced a SegNet selection method to combat feature overload through segmentation, using CNNs for image classification and focusing on reducing feature maps to decrease drone image processing time and enhance water body detection capabilities.

In water body segmentation, the extraction of global features, multiscale features, and detailed information is crucial. Global features reflect the overall characteristics of the target. Sui et al. [40] considered global relationships, such as repetitive textures in remote sensing images and incorporated a nonlocal sparse residual attention encoder into a residual dense network to capture similar contextual information from a global perspective. Huang et al. [42] proposed a transformer-based method that uses self-attention to learn globally relevant structural features, achieving segmentation on optical remote sensing images. Other networks utilizing transformer methods include Hi-ResNet [43] and LEFormer [44], which enhance the dynamic global modeling capability of transformers through multibranch modules stacked with information aggregation blocks. The progressive upsampling strategy proposed by Liu et al. [66] solves the feature compatibility problem of transformer-CNN, and Zhang et al. [67] develop a CNN-transformer hybrid generator for capturing local details and remote dependencies. For multiscale feature learning, the MKANet network [38] employs multibranch kernel-sharing atrous convolution to extract multiscale features from overhead remote sensing images, enhancing land cover classification performance. Tan et al. [30] recalibrated features extracted at different scales and adopted a top–down integration approach to enhance discriminative edge changes during segmentation of high-resolution images. Heidler et al. [34] combined semantic segmentation frameworks with edge detection frameworks for adaptive multiscale prediction, specifically extracting narrow coastal shorelines. Zhang et al. [35] utilized a lightweight dual-branch network that integrates attention modules and feature fusion modules to acquire contextual multiscale information. Liu et al. [36] used global thumbnails and positional masks to adaptively capture contextual information at different scales. Ovi et al. [31] learned multilevel features through deep supervision and extensively redesigned skip connections to reduce the impact of irrelevant features in the background, allowing for high-precision automatic extraction from high-resolution satellite images. Detail features are a type of characteristic that most models aim to learn. Wang et al. [23] divided the transformer decoder into two subdecoders to recognize the overall contour of the target and local details. The encoder utilizes depth-variable convolutional layers and residual learning blocks to extract and retain geometric detail information from SAR images as much as possible [24]. To obtain high-resolution images with texture details for segmentation, Sui et al. [41] proposed a diffusion model with an adversarial learning strategy to enhance the detailed texture prediction of the diffusion model.

In addition to the aforementioned common feature learning approaches, Van Soesbergen et al. [37] performed point-level feature extraction for reservoirs to enhance the similarity among water body points and the dissimilarity between different land covers. NFANet [39], on the other hand, employs a weakly supervised method with adjacent samplers, capturing neighboring features rather than global or local features. Based on these research methodologies, we designed modules focused on learning both global and local features, as well as fusion modules to meet the extraction of multilevel features. Additionally, we incorporated NIR bands and NDWI information to assist in water body extraction.

III. Methodology

In this article, to address the challenge of complex waterbody extraction, we propose a novel remote sensing-based

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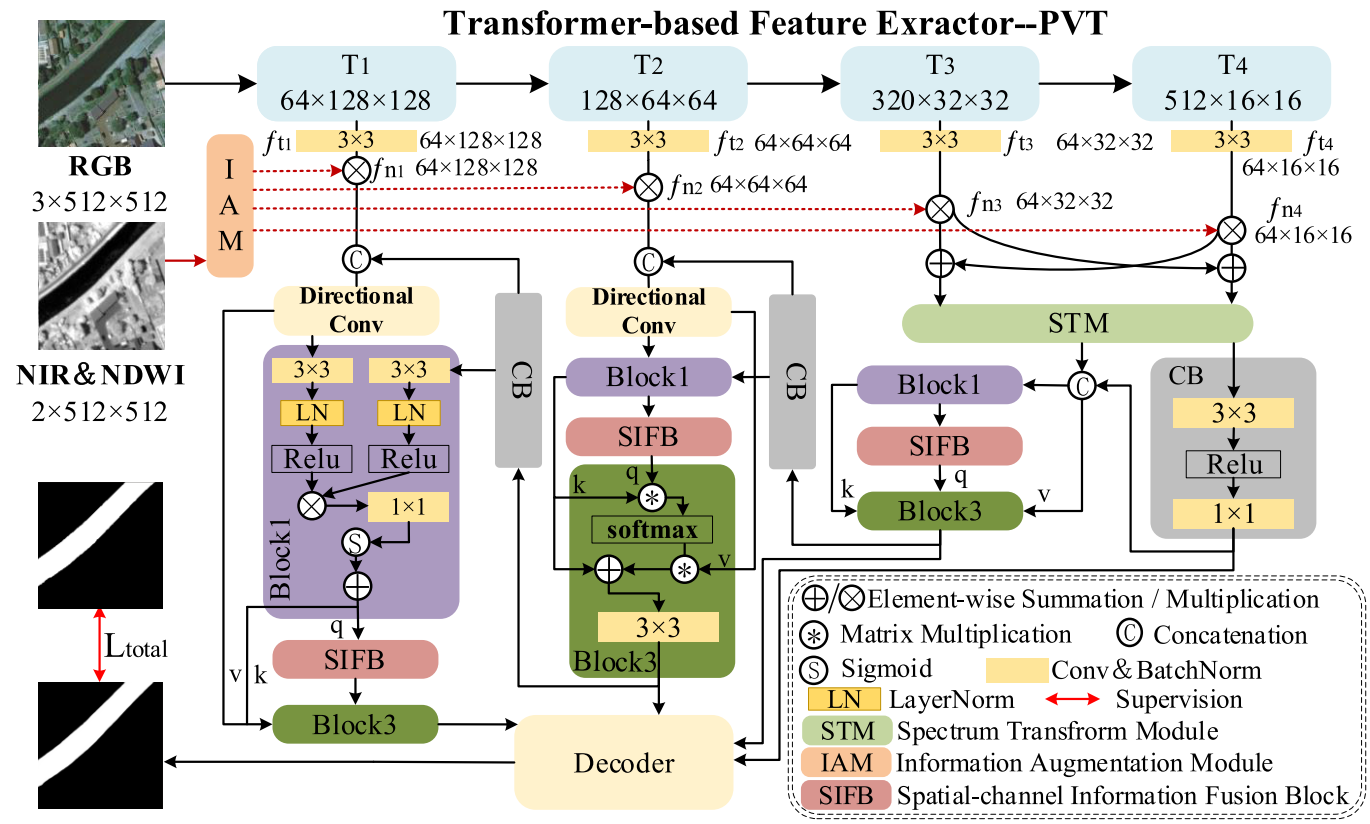


Fig. 2. Overall architecture of the proposed transformer-driven CWBSNet.

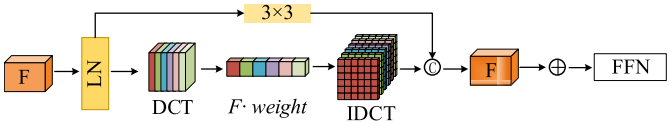


Fig. 3. Proposed architecture of the STM.

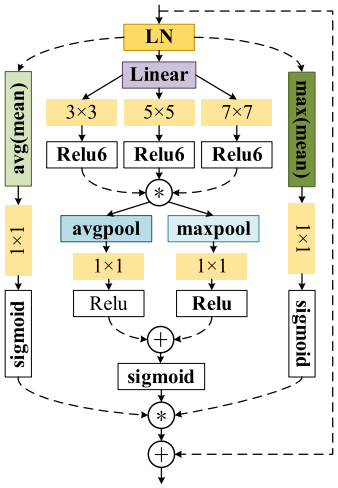


Fig. 4. SIFB of MFEFM.

network, termed CWBSNet. The overall architecture is illustrated in Fig. 2. The core innovation of CWBSNet lies in the introduction of a waterbody-aware encoder, which is composed of three key modules: spectral-domain transformation module, multiscale feature enhancement and fusion module, and interactive attention mechanism (IAM). The MFEFM is further subdivided into Block1, a spatial-channel information fusion block (SIFB), and Block3, aiming to refine multilevel feature representations. The input imagery is divided into two groups: RGB bands and NIR-NDWI bands. The RGB images are processed by a backbone network—pyramid vision transformer (PVT)—to extract a feature set ft. Meanwhile, the NIR-NDWI pair is processed by the IAM module to extract features fn, fully leveraging the strong water absorption characteristics of the NIR band and the edge sensitivity of the NDWI index. These two feature sets are then deeply fused via a feature fusion layer and fed into the waterbody-aware encoder.

Within the encoder, the deep features of size 16 × 16 are first passed through the STM module, which enhances the model's capacity for learning localized features and capturing semantic representations of waterbodies. Subsequently, features are upsampled by the contextual block (CB), which progressively propagates multiscale information and facilitates global contextual modeling. At each hierarchical level, features are further processed by the MFEFM module, which performs compression and reconstruction to better capture fine details. Notably, before feeding the shallow features (128 × 128 and 64 × 64) into MFEFM, we introduce directional convolution to strengthen the model's perception of orientation-specific

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structures and better delineate the complex boundaries of waterbodies.

Finally, multilevel features are decoded into predicted segmentation masks. Supervised by ground-truth labels, the model is optimized using a hybrid loss that combines cross-entropy loss and IoU loss, which further boosts segmentation accuracy.

Throughout the segmentation pipeline, the STM, MFEFM, and IAM modules are designed to address specific challenges, such as variable waterbody reflectance, complex boundary shapes, and texture diversity. Each module operates on feature maps at different levels. Specifically, STM enhances the model's robustness to reflectance variation through joint spectral-texture modeling; MFEFM captures fine-scale details via multiscale feature fusion and reconstruction; and IAM strengthens the contribution of NIR-NDWI bands through an attention-based mechanism.

In the following sections, we will be detailed to describe the four main components of CWBSNet: Section III-A the design of STM. Section III-B the mechanism of the MFEFM module. Section III-C the specific implementation of IAM. Section III-D loss function strategy. Comprehensively elaborate on the technical implementation and innovation of our model.

A. Spectrum Transform Module

To address the issue of waterbody discontinuity caused by abrupt changes in reflectance, this article proposes the spectrum-texture module. The STM module is specifically shown in Fig. 3. The core idea of STM is to enhance the model's robustness to reflectance variation through joint modeling in the frequency and spatial domains. STM receives deep features $ft4(16 \times 16)$ and $ft3(32 \times 32)$, which are first fused and normalized. These fused features are then passed through convolutional layers and discrete cosine transform to separate frequency-domain representations of water and adjacent land surfaces. The resulting frequency components are weighted using trainable weight matrices, followed by inverse DCT to reconstruct spatial-domain features, thereby preserving local reflectance consistency and mitigating segmentation discontinuities.

Subsequently, a feed-forward network (FFN) based on the transformer architecture extracts both local and global features through fully connected layers and deep convolutions, enriching semantic feature representations. The innovations of STM can be summarized as follows: 1) joint frequency–spatial domain modeling effectively addresses abrupt spectral changes; 2) multilevel feature fusion improves the capture of waterbody structures and semantic information; and 3) integration of transformer-based FFN enhances global contextual modeling.

This module significantly improves the segmentation accuracy of complex waterbodies and lays a solid foundation for subsequent multiscale feature decoding.

In the context of discrete cosine transform applications, the STM module demonstrates clear novelty and advantages in both technical design and performance. Traditional DCT-based methods are often limited to single-domain feature processing and struggle to capture global spectral patterns alongside local spatial details, resulting in insufficient adaptability to reflectance variations. STM addresses these limitations by introducing a frequency–spatial co-modeling mechanism: it transforms deep features into the frequency domain using DCT, dynamically adjusts the frequency responses with trainable weight matrices, and reconstructs the spatial features through IDCT. This bidirectional modeling not only retains the global spectral information captured in the frequency domain but also enhances the local consistency in the spatial domain, significantly improving segmentation continuity and overcoming the spatial representation limitations of conventional DCT approaches.

Moreover, by integrating a transformer-based FFN, STM strengthens its ability to model global contextual dependencies, enabling the network to better understand the complex relationships between waterbodies and their surrounding environments.

Although the DCT has been widely used in frequency-domain analysis, the existing DCT-based approaches are mostly static transformations that lack learnability, making them less effective in adapting to the complex spectral variations present in remote sensing imagery. In contrast, the proposed STM integrates DCT with spatial convolution and a trainable weighting matrix to enable dynamic modulation of spectral features, thereby enhancing the model's ability to handle abrupt changes in edge reflectance.

1) *Learnable Frequency-Domain Weighting Mechanism:* The trainable weight matrix adjusts the contributions of different frequency components during training, allowing the model to selectively focus on spectral features relevant to water boundaries based on contextual information.
2) *IDCT-Based Spatial Reconstruction:* After frequency-domain processing, STM reconstructs the spatial features via the IDCT, ensuring the module's backpropagation compatibility and end-to-end trainability—setting it apart from traditional DCT methods that function as static, preprocessing steps.
3) *Seamless Integration With Transformer Architecture:* STM is embedded within the FFN and works in synergy with the attention mechanism, facilitating hierarchical spectral modeling. This design introduces a novel fusion of frequency-domain analysis and deep attention-based learning.

These characteristics overcome the limitations of traditional DCT-based methods—namely, being static, nonlearnable, and context-independent—and offer a new pathway for spectral modeling in complex scenarios. To the best of our knowledge, there is currently no prior work employing such a frequency-attention co-design for water body segmentation tasks.

B. Multilevel Feature Extraction and Fusion Module

In remote sensing image waterbody extraction tasks, blurred waterbody boundaries often result from fine boundary details being overwhelmed by noise. Moreover, insufficient multiscale feature extraction prevents models from simultaneously capturing both global waterbody structures and local boundary

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details, significantly degrading segmentation accuracy and robustness—especially in complex scenes.

To address these challenges, we propose the multiscale feature enhancement and fusion module. This module is designed to enhance the model's perception of waterbody boundary details through hierarchical feature extraction and attention mechanisms. The necessity of MFEFM is reflected in the following aspects.

First, the issue of boundary blurring is fundamentally caused by inadequate multiscale feature learning. To this end, Block1 utilizes multiscale convolution and feature fusion to capture the hierarchical information of local reflectance and boundary details of waterbodies.

Second, due to the spectral similarity between water and adjacent land areas, the model must possess strong contextual modeling capability. Therefore, Block3 introduces a self-attention mechanism to compute attention weights between waterbody textures and nearby land features, enhancing global contextual associations.

Finally, to improve the model's sensitivity to fine details, the intermediate SIFB employs depthwise separable convolutions to extract multiscale features. As shown in Fig. 4, it integrates adaptive pooling and channel attention to deeply fuse channel and spatial information. In addition, residual connections are used to preserve input information and prevent feature degradation.

By combining multiscale feature extraction, attention mechanisms, and spatial–channel fusion, the MFEFM significantly enhances the model's ability to learn fine boundary details. This effectively mitigates segmentation accuracy drops caused by boundary ambiguity, providing a robust and efficient solution for complex waterbody extraction tasks.

C. Information Augmentation Module

To better address the complexity and diversity of waterbodies, we propose the IAM, which is designed to fully exploit the unique sensitivity of the NIR band and the NDWI to water. By incorporating an improved linear attention mechanism, IAM enhances the model's ability to represent complex waterbody features. The design principles of the IAM module are shown in Fig. 5.

First, the NIR band exhibits strong absorption characteristics for water, and the NDWI is highly sensitive to water boundaries, providing crucial spectral cues for waterbody extraction—cues that conventional RGB bands often fail to capture. IAM explicitly processes the NIR-NDWI combination to significantly improve the model's spectral discrimination capability for water.

Second, conventional transformer architectures suffer from high computational complexity, making them inefficient for high-resolution remote sensing images. To address this, IAM adopts an improved rotary position embedding (RoPE)-based linear attention, which encodes relative positional information in multidimensional space. This enhances the model's ability to learn spatial distributions of waterbodies efficiently.

Third, to capture multiscale features of waterbodies, IAM integrates deep convolutional layers, linear projections, activation functions, and gating mechanisms, in combination with multihead self-attention to re-establish global dependencies among different spatial locations. Layer normalization and residual connections are employed to ensure model stability and convergence.

IAM focuses specifically on features from the NIR and NDWI bands, while the PVT backbone extracts RGB features. The fusion of these complementary sources leverages the strengths of each modality and enables cooperative processing of multisource data. The advantages of the IAM module can be summarized as follows.

Spectral Complementarity: The NIR-NDWI combination offers strong absorption contrast and boundary sensitivity for water in the infrared range, while RGB bands provide rich color and texture information in the visible spectrum. Their fusion enables comprehensive waterbody representation.

Contextual Enhancement: The improved linear attention mechanism in IAM captures long-range dependencies and global context related to water distribution, which, when combined with local RGB details, improves the accuracy of boundary recognition, especially for complex waterbodies.

Multiscale Feature Fusion: The multiscale feature extraction capability of IAM, combined with the hierarchical representations inherent to transformer architectures, achieves multilevel fusion from local to global scales, significantly enhancing segmentation performance.

By explicitly processing the NIR-NDWI inputs and fusing them with RGB features through an enhanced linear attention mechanism, the IAM module greatly improves the model's ability to represent and segment complex and diverse waterbodies.

D. Loss

In the task of complex waterbody segmentation, a composite loss function is employed between the ground truth (GT) labels and the predicted results to achieve high-precision segmentation. This loss function consists of two components: binary cross-entropy (BCE) loss and intersection over union (IoU) loss. The BCE loss focuses on optimizing pixel-level classification of water regions, while the IoU loss enhances the overall overlap between predicted and actual waterbodies. The combination of the two helps improve boundary segmentation accuracy and balances the optimization of both local and global features. The definition of the BCE loss is given in the following equation:

$$L_{\mathrm{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \cdot \log(\sigma(p_i)) + (1-y_i)\cdot \log(1-\sigma(p_i))\right] \tag{1}$$

where p_i is the predicted probability of the ith pixel, y_i is the corresponding GT label, and σ denotes the sigmoid function. The definition of the IoU loss is provided in the following equation:

$$L_{\mathrm{IoU}} = 1 - \frac{\sum(p_i \cdot y_i)}{\sum p_i + \sum y_i - \sum(p_i \cdot y_i)} \tag{2}$$

where p_i and y_i represent the predicted probability and the GT label, respectively. The final loss function is a weighted sum

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of these two components, as shown in the following equation:

$$L = L_{\text{BCE}} + L_{\text{IoU}}. \tag{3}$$

By combining these two loss functions, our model can improve the segmentation quality of waterbody boundaries and achieve a balance between local and global feature optimization. Moreover, in scenarios with an imbalance between foreground (water) and background pixels, this approach helps mitigate the model's bias toward the majority class.

IV. Experiments and Results

A. Experimental Settings

1) Datasets: To demonstrate the versatility of our approach for remote sensing imagery, we selected two water body datasets of different resolutions—S1S2-Water and FLAIR #1—as experimental subjects. These datasets allow us to assess the generalizability of our model and compare its performance with other algorithms. For each dataset, training and testing sets were split to facilitate an in-depth evaluation of strengths and weaknesses. The S1S2-Water dataset adheres to remote sensing benchmark guidelines and includes diverse climate, atmospheric, and land cover conditions across different seasons. Derived from Sentinel-2 imagery, it spans 29 countries with a total coverage area of 650 000 km^2. Each image measures 10 980 × 10 980 pixels at a spatial resolution of 10 m. In this study, we cropped 42 satellite images into 512 × 512 patches, resulting in a dataset containing 10 607 labeled water body images, of which 7992 were used for training and 2615 for testing. The FLAIR #1 dataset consists of Earth observation data from various aerospace sensors with a spatial resolution of 0.2 m. This dataset reflects real-world cases in land cover mapping tasks across different landscapes and climates within metropolitan France. It comprises 12 963 labeled water body images, each 512 × 512 pixels, of which 10 960 images were used for training and 2003 for testing.

2) Network Implementation: All experiments were conducted using PyTorch [17] on an NVIDIA GeForce RTX 3090 GPU with 24 GB of memory. To evaluate the efficiency and effectiveness of the CWBSNet method, we adopted the PVT-v2-b2 [18] as the backbone transformer encoder, initializing it with pretrained parameters to validate the effectiveness of the water body perception encoder. All newly added layers were initialized using the "Kaiming" method [19]. The loss function used was the sum of BCE loss and IoU loss. Model training utilized the Adam optimizer across 75 epochs with a batch size of 8 and an initial learning rate of 1e-4, decaying by a factor of 1/10 every 50 epochs.

3) Evaluation Metrics: We utilized several classical metrics commonly applied in assessing the quality of semantic segmentation model predictions, including precision (P), recall (R), $F1$-score ($F1$), and IoU metrics, as shown in (4)–(7). These metrics are calculated based on the confusion matrix. Generally, a higher P indicates fewer false positives, while a higher R reflects fewer false negatives. When considering overall model performance—especially in contexts where balancing false positives and false negatives is crucial—the $F1$, as the harmonic mean of P and R, serves as an important evaluation metric. IoU measures the ratio of the intersection of predicted water pixels and GT water pixels to their union. Among these metrics, the $F1$ and IoU are comprehensive indicators. We applied all four metrics to quantitatively evaluate all comparison methods as well as our own method across the two datasets

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{4}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{5}$$

$$F1\text{-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{6}$$

$$\text{IoU} = \frac{P \times R}{R - P \times R + P} = \frac{\text{TP}}{\text{FN} + \text{TP} + \text{FP}} \tag{7}$$

where TP represents the number of true positives, FP represents the number of false positives, and FN represents the number of false negatives.

B. Comparison With State of the Arts

To validate the effectiveness of the proposed CWBSNet, we present experimental results based on the S1S2-water Sentinel-2 water body dataset and the FLAIR #1 aerial water body dataset. We compared our method with several state-of-the-art models for semantic segmentation, including Rolling-Unet, SCTransNet, Swin-Unet, UNetFormer, RMT, GeleNet, Log-can, ACCoNet, Sacanet, DOCNet, and Logcan++.

1) Quantitative Comparison on the S1S2-Water and FLAIR #1 Datasets: In Table I, we outline the quantitative comparison results between our method and 11 other benchmark methods on the S1S2-Water and FLAIR #1 datasets. Based on the results, our method outperforms all other methods across both datasets, except in terms of R on the S1S2-Water dataset and P and $F1$ on the FLAIR #1 dataset. Specifically, on the S1S2-Water dataset, our method achieves a P score that is 9.01% higher than the second-best approach. Although our R score is 6.29% lower than the top-performing method, our $F1$-score and IoU score are higher than those of the second-best method by 5.05% and 1.51%, respectively. On the FLAIR #1 dataset, our P score is 3.58% below that of the leading method, and R is only 0.03% higher than the second-best method. However, our $F1$ trails the second method by only 0.83%, while our IoU score exceeds the second-best method by 2.35%.

In addition, we observed that the S1S2-Water dataset, due to its lower resolution, leads to lower accuracy in many segmentation algorithms. In contrast, on the aerial FLAIR #1 dataset, the performance of various comparison algorithms is relatively similar, and the overall detection results are better than those on the S1S2-Water dataset. However, our proposed CWBSNet demonstrates strong detection performance on both datasets, indicating that it is suitable for water body extraction from remote sensing images and can adapt to images with different resolutions.

2) Model Complexity and Time Complexity Analysis: To comprehensively evaluate the effectiveness of the proposed model, we adopt three metrics: the number of parameters (#Params) to measure model complexity, giga floating-point

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TABLE I

QUANTITATIVE COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON S1S2-WATER AND FLAIR #1 DATASETS. WE MARK THE TOP TWO RESULTS IN RED AND BLUE, RESPECTIVELY

Method	Params (M)	FLOPs (G)	S1S2-Water(10m)					FLAIR #1(0.2m)				
			T(h)	P(%)	R(%)	F1(%)	IoU(%)	T(h)	P(%)	R(%)	F1(%)	IoU(%)
Swin-Unet$_{22}$[48]	27.15	5.92	40.0	54.18	43.49	50.24	57.13	55.3	75.95	68.22	68.68	62.92
UNetFormer$_{22}$[49]	11.68	11.74	6.2	40.48	56.11	49.03	46.72	7.9	45.35	59.81	53.59	51.96
GeleNet$_{23}$[50]	27.58	25.04	9.2	67.81	78.50	70.03	73.41	13.5	78.20	87.24	79.79	73.62
Sacanet$_{23}$[51]	51.79	95.50	7.5	43.68	67.09	53.91	56.67	12.5	62.20	74.24	67.69	58.54
ACCONet$_{23}$[52]	102.55	737.99	23.4	49.39	79.31	54.23	60.80	32.7	65.97	88.93	71.65	63.74
Logcan$_{23}$[53]	54.27	111.97	10.0	57.80	63.67	58.52	68.31	12.5	76.10	74.52	72.69	67.57
RMT$_{24}$[54]	42.56	63.05	65.4	58.20	66.88	57.34	66.96	85.4	75.97	74.49	71.98	66.42
Rolling-Unet$_{24}$[55]	28.32	131.48	66.7	63.26	83.62	68.91	75.17	86.7	80.01	78.72	75.92	70.46
SCTransNet$_{24}$[56]	11.19	40.46	100.5	51.71	80.73	55.97	49.77	183.3	74.89	77.80	72.26	66.47
DocNet$_{24}$[57]	65.67	91.11	20.1	52.30	56.34	52.26	65.15	25.5	74.95	73.61	71.59	66.36
Logcanplus$_{24}$[58]	54.39	112.52	10.0	57.92	63.65	58.45	68.30	15.1	75.66	72.28	72.92	67.75
ours	30.33	59.60	43.1	76.82	77.33	75.08	76.68	31.2	76.43	88.96	78.96	75.97

operations per second (GFLOPs) to assess theoretical computational cost, and total training time to evaluate computational efficiency. GFLOPs are calculated under the same conditions for all methods, with the input batch size uniformly set to 1. We compare the model complexity and time complexity of all approaches using their default settings as reported in the original papers.

As shown in Table I, compared to other waterbody extraction methods, CWBSNet demonstrates satisfactory performance in terms of model parameters, computational complexity, and training time. This advantage is attributed to its efficient parallel computation optimization and modular design, enabling faster model training. Although CWBSNet is not the most lightweight model in terms of parameters or FLOPs, it achieves significantly higher detection accuracy than other methods with relatively low computational cost, while notably reducing training time. This balance between efficiency and performance makes CWBSNet well-suited for large-scale remote sensing image processing tasks.

The parameter settings for the comparison methods are consistent with the default settings in the original paper. As shown in Table I, compared to other water extraction methods, CWBSNet demonstrates satisfactory performance in both model parameters and time complexity. Although the proposed CWBSNet does not achieve the lowest model parameters or time complexity, it significantly outperforms other models in detection accuracy while consuming much less computational cost.

3) Visual Comparison: To make our comparison experimental results more visually accessible, we visualize the segmentation results, as shown in Figs. 6 and 7. In these figures, blue indicates TP, red represents FP, and green denotes FN. In the S1S2-Water dataset, to validate the effectiveness of the proposed method in addressing water body disconnection, boundary blurring, and missing detection, we selected images containing various water body shapes, meandering tributaries, and coexisting water bodies for a detailed comparison. The resulting saliency maps are visualized in the images, as shown in Fig. 6. Specifically, the first and second rows depict diversified water body coexistence scenarios, where small ponds and lakes have smaller catchment areas compared to rivers. Other methods tend to miss or incorrectly detect boundaries for these small water bodies. The third and fourth rows show remote sensing scenes with dense tributaries, and we find that only our method successfully detects all river branches and accurately extracts the boundaries, ensuring connectivity of the river water bodies. The fifth row presents large river detection scenes, where the remote sensing scenes are relatively simple, and many methods can sufficiently detect the water bodies. However, minor boundary misdetections occur in meandering river sections.

As shown in Fig. 7, for the FLAIR #1 dataset, where the resolution of the aerial imagery is noticeably higher than that of Sentinel-2, the detection of water bodies requires more precise extraction. We again selected a variety of representative remote sensing scenes. The first row shows diverse water body images, including rivers mixed within urban areas. Other algorithms tend to misdetect at the river junctions and along the river edges, and irregular water bodies are often missed. The second, fifth, and sixth rows depict large water body

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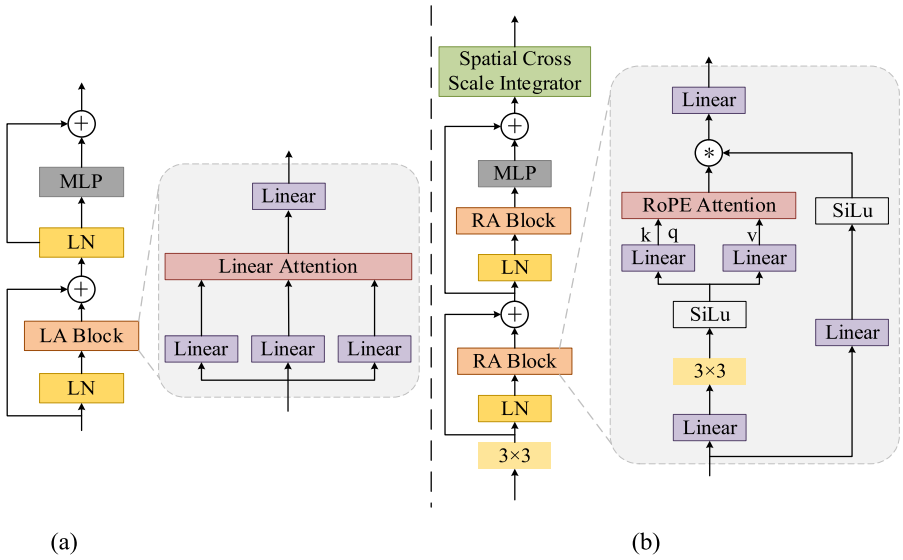


Fig. 5. Left figure (a) shows the architecture of the linear attention transformer, and the right figure (b) shows the information enhancement module we proposed.

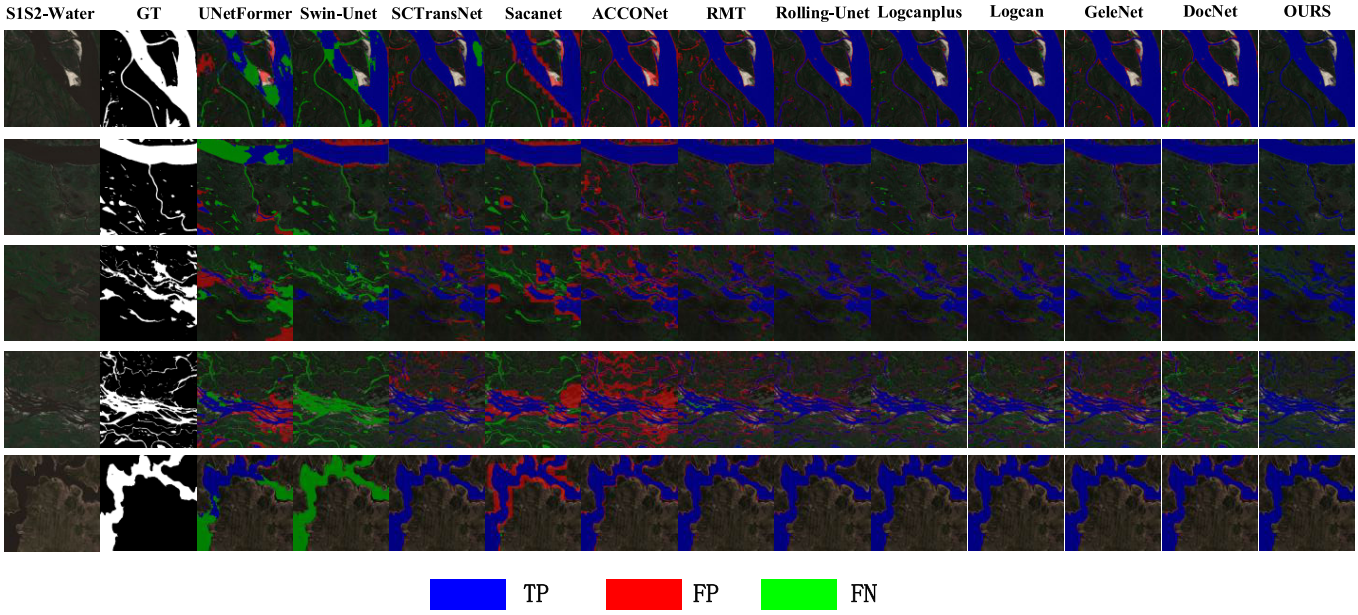


Fig. 6. Visual comparison of different methods on S1S2-Water dataset.

scenes, where our method achieves the best extraction results. The fourth row shows a river obstructed by trees along the riverbanks, causing boundary blurring and similar features to surrounding roads. Other methods misdetect the road as water and fail to correctly identify the water body boundaries in obstructed areas. While our method also suffers from some misdetections, it still produces the most accurate detection results overall.

C. Ablation Studies

We conducted a comprehensive ablation study on the S1S2-Water dataset to evaluate the effectiveness of each module in our CWBSNet. For a more detailed analysis, we have shown eight typical and challenging remote sensing water body scenarios in the figure, allowing for a more intuitive understanding of the ablation study for each module. These scenarios include irregular water bodies with blurred boundaries, water bodies with clear boundaries but varying reflectance, major rivers with high flow, smaller tributary rivers with low flow, and small lakes with diverse shapes. The ablation studies were carried out using the PVT-v2-b2 backbone in CWBSNet, with the same parameter settings and dataset splits, as described in Section I. In addition, we have presented the visual effects after the addition of NIR and NDWI, clearly demonstrating the utility of the IAM module and the additional information that the NIR and NDWI bands can provide.

1) Individual Contribution of Three Modules: To analyze the performance of the STM, MFEFM, and IAM modules, we designed various combinations of the three modules,

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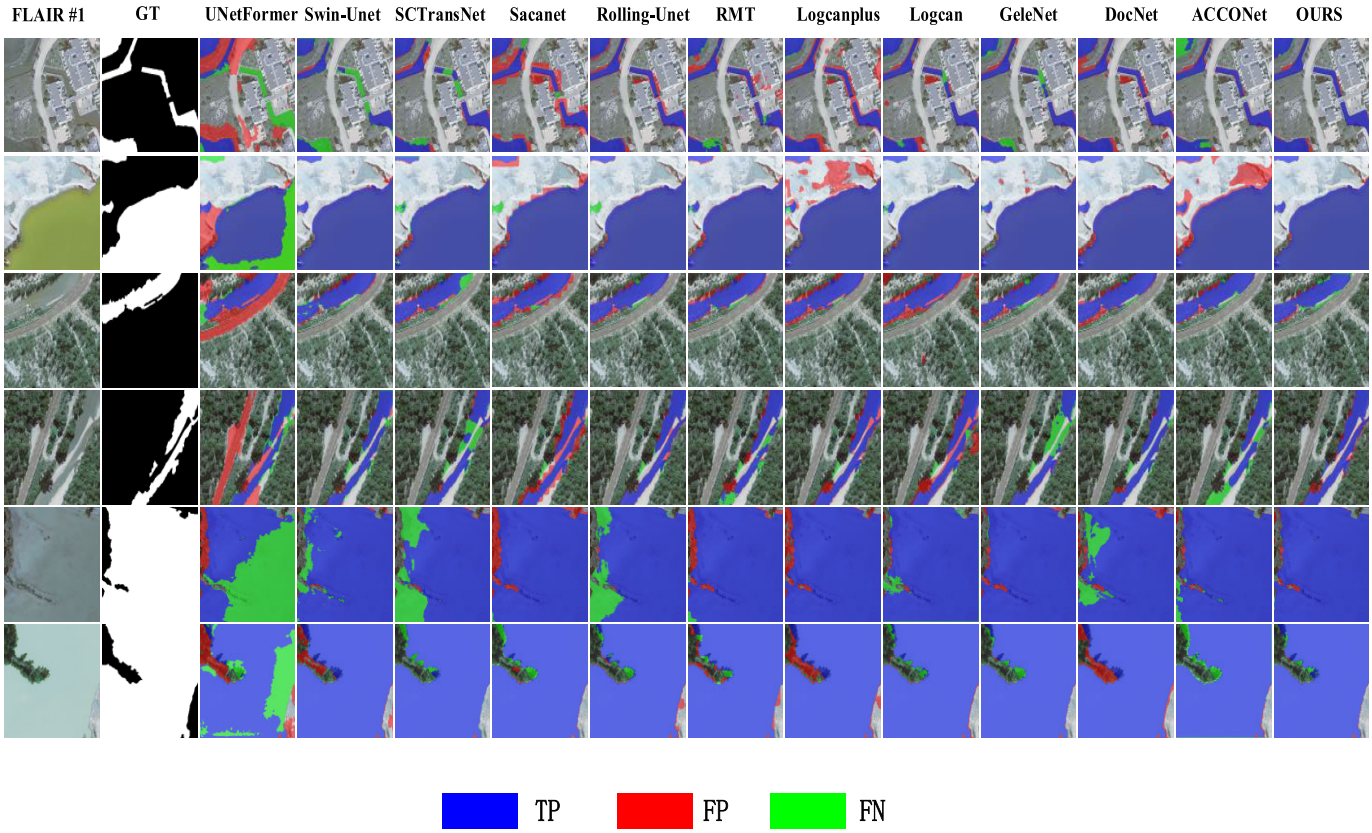


Fig. 7. Visual comparison of different methods on FLAIR #1 dataset.

TABLE II
ABLATION RESULTS OF EVALUATING THE INDIVIDUAL CONTRIBUTION OF EACH MODULE IN CWBSNET. THE BEST ONE IN EACH COLUMN IS BOLD

No.	Baseline	STM	MFEFM	IAM	Test			
					P	*R*	*F1*	*IoU*
1	√				0.7332	0.7196	0.6884	0.7323
2	√	√			0.7169	0.7672	0.7176	0.7474
3	√		√		0.7260	0.7631	0.7203	0.7473
4	√			√	0.7026	0.7751	0.7121	0.7436
5	√	√	√		0.7238	0.7831	0.7303	0.7539
6	√	√		√	0.7076	**0.8276**	0.7393	0.7545
7	√		√	√	0.7408	0.7788	0.7365	0.7582
8	√	√	√	√	**0.7682**	0.7733	**0.7508**	**0.7669**

resulting in eight variants: 1) baseline, where we removed all proposed modules and directly output the four features ft_1, ft_2, ft_3 and ft_4, they obtained from the PVT backbone to the decoder for prediction; 2) baseline + STM; 3) baseline + MFEFM; 4) baseline + IAM; 5) baseline + STM + MFEFM; 6) baseline + STM + IAM; 7) baseline + MFEFM + IAM; and 8) baseline + STM + MFEFM + IAM, which is our CWBSNet model. Quantitative experiments were conducted on the S1S2-Water dataset, and the results of the experiments are reported in Table II.

From Table II, examining rows one to five, we observe that, in the test set, while each module led to a slight decline in P, they individually boosted R by approximately 4%, $F1$-score by around 2.8%, and IoU by about 1.1% over the baseline. This suggests the effectiveness of the three proposed modules. Rows six to eight in Table II present the results of pairwise module combinations. Compared to the first few rows, P fluctuated by about 2% relative to the Baseline, while R increased by approximately 3%, $F1$ by around 2%, and IoU by about 1.1%. These findings indicate that collaboration between different modules further enhances the robustness of the algorithm, resulting in improved performance. Therefore, when all three modules work together, our method achieves optimal P. Although R is slightly lower, the comprehensive

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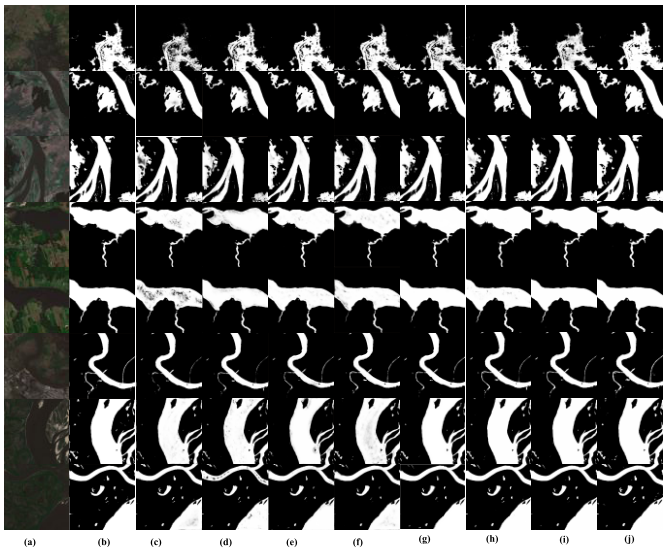


Fig. 8. Visual comparison of individual addition of STM, MFEFM, and IAM modules as well as pairwise adaptation and final results on S1S2-Water dataset. (a) Images. (b) GT. (c) Baseline. (d) Baseline +STM. (e) Baseline +MFEFM. (f) Baseline +IAM. (g) Baseline +STM+MFEFM. (h) Baseline +STM+IAM. (i) Baseline +MFEFM+IAM. (j) Ours.

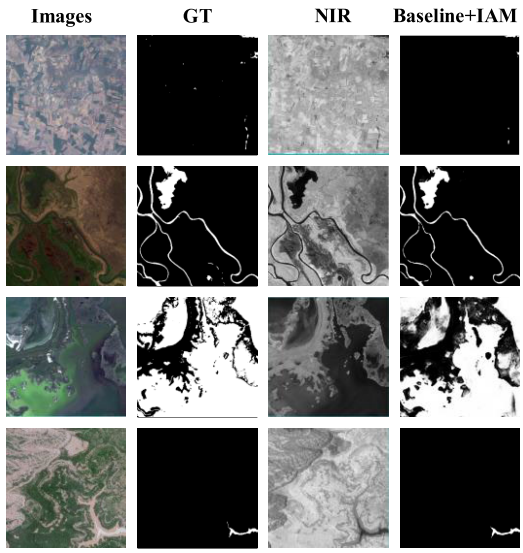


Fig. 9. Visual comparison of the NIR and IAM modules was conducted on the S1S2-Water dataset.

metrics $F1$ and IoU outperform the baseline by 6.2% and 3.4%, respectively.

2) Visual Presentation of Three Modules: In Test1 and Test2, we selected four images each to further illustrate the effects of our ablation study under diverse water bodies and complex backgrounds, as shown in Fig. 8. Without any modules, the baseline method yielded suboptimal saliency maps, exhibiting issues, such as inaccurate edge extraction, incomplete highlighting, and omissions in narrow river segments. By incorporating the STM, which captures edge detail, the "Baseline + STM" prediction maps successfully highlighted edge details, reducing inaccuracies in edge localization (evident in the first three images). However, with the increased focus on edge details, there was a tradeoff as the main water body regions appeared less distinct, showing incomplete highlighting. The "Baseline + MFEFM" setup demonstrated the MFEFM ability to capture global water features, achieving a more distinct highlighting of the water regions (especially visible in the third and fourth images), although omissions occurred in narrow water streams. Meanwhile, IAM, which leverages NIR and NDWI spectral information to differentiate water from diverse remote sensing backgrounds, accurately highlighted the water bodies, particularly narrow and diverse waterways, in the "Baseline + IAM" maps. As expected, combining modules, as seen in Fig. 8, integrated the strengths of each individual module. With all three modules operating together, our model produced saliency maps that visually resemble the GT the closest. This analysis confirms the effectiveness of the three proposed modules, demonstrating that each module fulfills a distinct function without conflicts, leading to enhanced overall model performance.

3) Visual Presentation of NIR and NDWI: We selected four Sentinel-2 images with significantly different spectral characteristics, as illustrated in Fig. 9. The water bodies in these images exhibit notable heterogeneity in terms of spectral reflectance, geometric shape, and spatial distribution. In the experiments, we present the original multispectral images, GT labels, NIR band images, and the detection results based on the Baseline + IAM module.

Experimental analysis demonstrates the unique physical advantages of the NIR band in water body detection: water bodies exhibit strong absorption in the NIR band, with reflectance values significantly lower than other land cover types, appearing as dark or black regions in the imagery. In contrast, vegetation, soil, and man-made structures typically have higher reflectance in the NIR band, creating a clear spectral separability between water and nonwater classes. This provides a reliable physical basis for accurate segmentation and extraction of water bodies.

By enhancing spectral contrast between land cover types, the NIR band significantly improves classification accuracy and robustness in water body detection. Although the segmentation results of the Baseline + IAM module show some local deviations compared to the GT labels, the module accurately captures the spatial distribution patterns of water regions and achieves high spatial coherence (IoU) with the GT. This confirms the effectiveness and practicality of the module in complex water body detection tasks.

To evaluate the effectiveness of the NIR and NDWI bands, we designed several input configurations with different combinations of spectral information, resulting in four variants: 1) a baseline model without the IAM module, i.e., without any additional NIR or NDWI information; 2) IAM module with only NIR input; 3) IAM module with only NDWI input; and 4) our full CWBSNet network. Quantitative experiments were conducted on the S1S2-Water dataset, and the results are reported in Table II.

In the first row of Table III, where no additional spectral information is provided, the model achieves the lowest performance, with an $F1$-score of 73.03% and an IoU of 75.39%. After incorporating the NIR band through the IAM module, the $F1$-score and IoU increase by 1.53% and 0.73%,

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TABLE III
ABLATION RESULTS OF EVALUATING THE INDIVIDUAL CONTRIBUTION OF EACH BAND IN CWBSNET

No.	NIR	NDWI	Test			
			P	*R*	*F1*	*IoU*
1	×	×	0.7238	0.7831	0.7303	0.7539
2	√	×	0.7429	0.7763	0.7456	0.7612
3	×	√	0.7396	0.7756	0.7377	0.7581
4	√	√	0.7682	0.7733	0.7508	0.7669

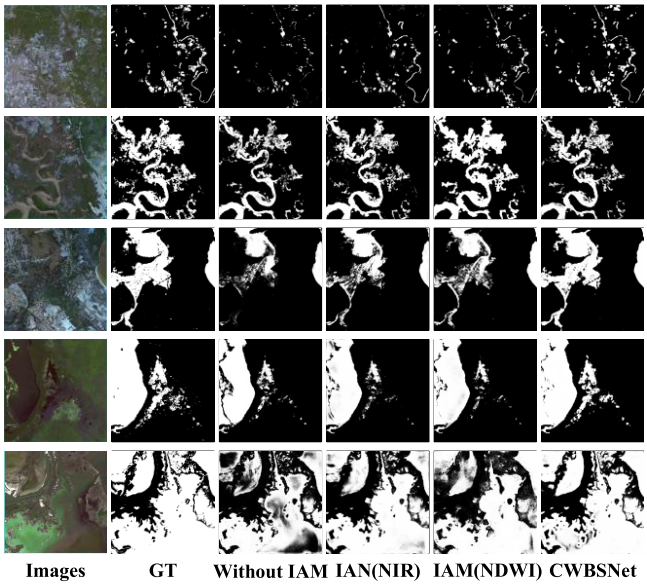


Fig. 10. Visual comparisons of the NIR and NDWI bands were conducted on the S1S2-Water dataset.

respectively. When using the NDWI band alone, the improvements are 0.74% and 0.42%, indicating that both spectral bands provide auxiliary benefits for water body extraction. Notably, when our CWBSNet leverages the IAM module to jointly learn from both NIR and NDWI bands, the $F1$-score and IoU further improve by 2.05% and 1.3%, respectively, demonstrating the practicality and effectiveness of integrating these spectral sources.

In addition, the experimental results in Table III are visualized in Fig. 10. Under the challenging and cluttered remote sensing scenarios of the S1S2-Water dataset—where water bodies exhibit irregular shapes and bear strong spectral similarities to surrounding land covers—the segmentation results without the IAM module (as shown in the third column) differ significantly from the GT, with large areas of missed segmentation. After incorporating NDWI and NIR individually, the model partially compensates for these omissions. However, the best performance is achieved when both NDWI and NIR are jointly utilized—i.e., in the CWBSNet—whose segmentation results are the closest to the GT.

V. CONCLUSION

To address the challenge of extracting complex water bodies in remote sensing images, we propose the complex water body network. CWBSNet introduces the STM, multiscale feature extraction and fusion module, and infrared-assisted module to refine water boundary details, capture global water features, and leverage NIR and NDWI for auxiliary learning. This combination effectively enhances the model's ability to perceive diverse and complex water bodies without compromising traditional water extraction accuracy. Our CWBSNet algorithm presents broad applicability in remote sensing water body extraction, addressing the challenge of detecting diverse water bodies with high precision. Capturing varied water shapes enables better extraction of complex water structures, while designing feature learning methods tailored to specific targets yields improved outcomes. We also observe that for narrow rivers, low-resolution remote sensing images lack the pixel-level detail of high-resolution images, impacting postsegmentation water connectivity. Consequently, our future work will focus on extracting narrow rivers in lower resolution imagery.

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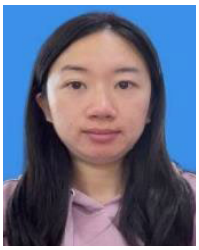
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